

49. Since the month begins with a Sunday, so there will be five Sundays in the month.

$$\therefore \text{Required average} = \left(\frac{510 \times 5 + 240 \times 25}{30} \right) = \frac{8550}{30} = 285.$$

50. Required average = $\left(\frac{32 \times 14 + 28 \times 13}{32 + 28} \right) = \left(\frac{448 + 364}{60} \right) = \frac{812}{60}$
 $= 13.53.$

51. Required average = $\left(\frac{50.25 \times 16 + 45.15 \times 8}{16 + 8} \right)$
 $= \left(\frac{804 + 361.20}{24} \right) = \frac{1165.20}{24} = 48.55.$

52. Required mean = $\left(\frac{60 \times 5 + 30 \times 10 + 20 \times 15}{5 + 10 + 15} \right)$
 $= \left(\frac{300 + 300 + 300}{30} \right) = \frac{900}{30} = 30.$

53. Total yearly income = ₹ $(5000 \times 5 + 5400 \times 7 + 2300)$
 $= ₹ (25000 + 37800 + 2300) = ₹ 65100.$

$$\therefore \text{Average monthly income} = ₹ \left(\frac{65100}{12} \right) = ₹ 5425.$$

54. Average marks = $\left(\frac{50 \times 40 + 60 \times 35 + 55 \times 45 + 45 \times 42}{40 + 35 + 45 + 42} \right)$
 $= \left(\frac{2000 + 2100 + 2475 + 1890}{162} \right)$
 $= \left(\frac{8465}{162} \right) = 52.25.$

55. Sum of m numbers = mn^2 . Sum of n numbers = nm^2 .
 Sum of $(m + n)$ numbers = $mn^2 + nm^2 = mn(m + n).$

$$\therefore \text{Average of } (m + n) \text{ numbers} = \frac{mn(m + n)}{(m + n)} = mn.$$

56. Let the five consecutive numbers be $z, z + 1, z + 2, z + 3$ and $z + 4$. Then,

$$\frac{z + (z + 1) + (z + 2) + (z + 3) + (z + 4)}{5} = x$$

$$\Rightarrow 5z + 10 = 5x$$

$$\Rightarrow z = \frac{5x - 10}{5} = x - 2.$$

So, the numbers are $x - 2, x - 1, x, x + 1, x + 2$.

$\therefore$ Required mean

$$= \frac{(x - 2) + (x - 1) + x + (x + 1) + (x + 2) + (x + 3) + (x + 4)}{7}$$

$$= \frac{7x + 7}{7} = x + 1.$$

57. Total quantity purchased = 4 kg.

$$\text{Total money paid} = ₹ \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) = ₹ \frac{25}{12}.$$

$$\therefore \text{Required average} = \left(4 \times \frac{12}{25} \right) \text{ kg/rupee}$$

$$= \left(\frac{48}{25} \right) \text{ kg/rupee} = 1.92 \text{ kg/rupee}.$$

58. Number of units of work completed by the five workers in 1 minute:

$$A \rightarrow \frac{1}{4}, B \rightarrow \frac{1}{5}, C \rightarrow \frac{1}{6}, D \rightarrow \frac{1}{10}, E \rightarrow \frac{1}{12}.$$

$$\therefore \text{Required average} = \left(\frac{\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{10} + \frac{1}{12}}{5} \right) = \left(\frac{\frac{4}{5} \times \frac{1}{5}}{5} \right) = \frac{4}{25}$$

$$= 0.16.$$

59. Total quantity of petrol consumed in 3 years

$$= \left(\frac{6460}{17} + \frac{6460}{19} + \frac{6460}{20} \right) \text{ litres}$$

$$= (380 + 340 + 323) \text{ litres} = 1043 \text{ litres}.$$

$$\text{Total amount spent} = ₹ (3 \times 6460) = ₹ 19380.$$

$$\therefore \text{Average cost} = ₹ \left(\frac{19380}{1043} \right) = ₹ 18.58.$$

60. Clearly, we have: $x = \left(\frac{3y + 3z}{6} \right)$ or $2x = y + z$.

61. Let the average expenditure be ₹ x . Then,

$$9x = 8 \times 30 + (x + 20) \text{ or } 9x = x + 260 \text{ or } 8x = 260$$

$$\text{or } x = 32.50.$$

$$\therefore \text{Total money spent} = 9x = ₹ (9 \times 32.50) = ₹ 292.50.$$

62. Let the original expenditure of the mess per day be ₹ x .

Then, new expenditure = ₹ $(x + 42)$.

$$\therefore \frac{x}{35} - \frac{(x + 42)}{42} = 1$$

$$\Rightarrow 6x - 5(x + 42) = 210$$

$$\Rightarrow x - 210 = 210$$

$$\Rightarrow x = 420.$$

63. Let their prices be $3x, 5x$ and $7x$ respectively.

$$\text{Then, } 3x + 5x + 7x = 15000 \times 3$$

$$\Rightarrow 15x = 45000$$

$$\Rightarrow x = 3000.$$

$$\therefore \text{Cost of cheapest item} = ₹ (3000 \times 3) = ₹ 9000.$$

64. Let the fourth number be x .

$$\text{Then, third number} = 5x, \text{ second number} = \frac{5x}{3} \text{ and first}$$

$$\text{number} = \frac{10x}{3}.$$

$$x + 5x + \frac{5x}{3} + \frac{10x}{3} = (24.75 \times 4) \text{ or } 11x = 99 \text{ or } x = 9.$$

$$\text{So, the numbers are } 9, 45, 15 \text{ and } 30.$$

$$\therefore \text{Largest number} = 45.$$

65. Let the second number be x . Then, first number = $2x$, third number = $4x$.

$$\therefore 2x + x + 4x = 56 \times 3 \Rightarrow 7x = 168 \Rightarrow x = 24.$$

$$\therefore \text{Required difference} = 4x - 2x = 2x = 48.$$

66. Let the third number be x . Then, second number = $2x$. First number = $4x$.

$$\therefore \frac{1}{x} + \frac{1}{2x} + \frac{1}{4x} = \left(\frac{7}{72} \times 3 \right) \text{ or } \frac{7}{4x} = \frac{7}{24} \text{ or } 4x = 24 \text{ or } x = 6.$$

$$\text{So, the numbers are } 24, 12 \text{ and } 6.$$